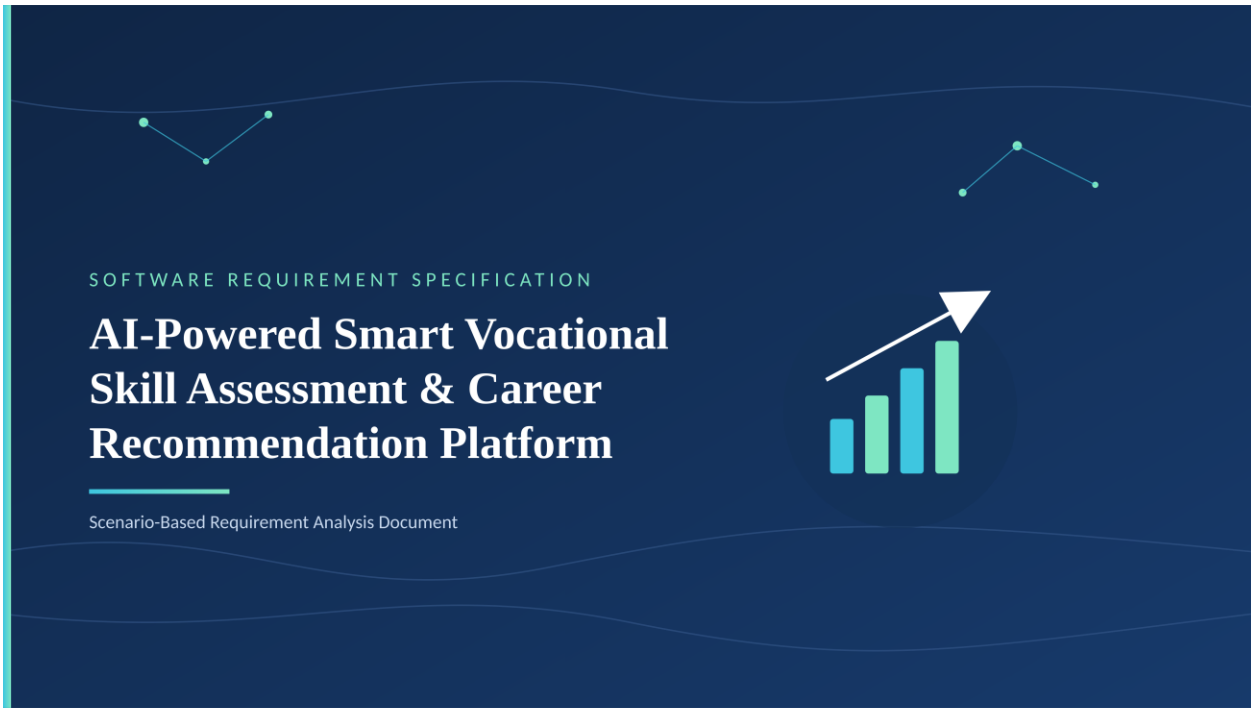
# CO2 ASSESSMENT TOOL 1 — SCENARIO-BASED REQUIREMENT ANALYSIS


## SIMATS ENGINEERING

*(Saveetha School of Engineering)*

Department of Computer Science and Engineering


| | |
|---|---|
| **STUDENT NAME** | **Nithiyasri S** |
| **REGISTER NUMBER** | **192572082** |
| **COURSE CODE** | **CSA1017** |
| **COURSE NAME** | **Software Engineering** |

# Table of Contents

# Problem Statement

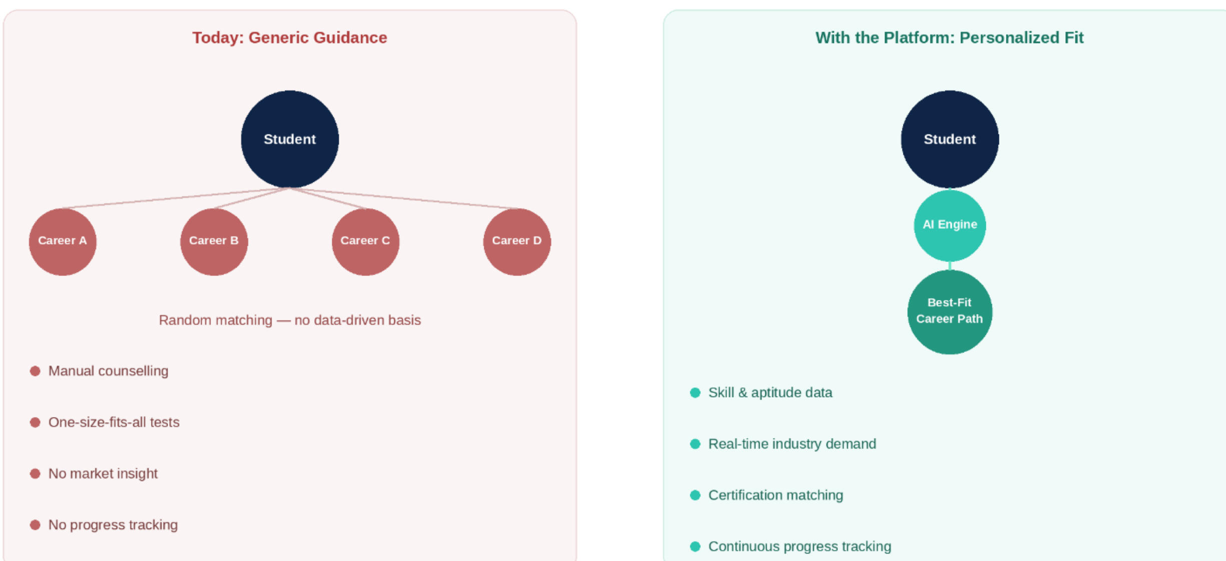
## 1. AI-Powered Smart Vocational Skill Assessment and Career Recommendation Platform

### Industry Context

Educational institutions, training organizations, and industries seek skilled professionals who match evolving job market requirements. AI, data analytics, and online assessment platforms help evaluate learners' competencies and recommend suitable career pathways and training programs.

### Problem Statement

Many students and job seekers struggle to identify careers that align with their skills and interests due to limited career guidance and generic assessments. Traditional career counseling methods often lack personalization and labor market insights. An AI-powered career recommendation platform should assess technical and soft skills, analyze aptitude and learning preferences, compare competencies with industry requirements, and recommend personalized career paths, certification programs, and job opportunities. The system should bridge the gap between education and employment.



# 1. Introduction

## 1.1 Purpose

This Software Requirement Specification (SRS) document describes the functional and non-functional requirements for the AI-Powered Smart Vocational Skill Assessment and Career Recommendation Platform. The purpose of this document is to provide a clear, complete, and unambiguous description of the system to be developed, so that it can serve as a common reference for students, faculty evaluators, and future development teams. It defines the scope of the system, the requirements of its stakeholders, and the constraints under which the system must operate.

## 1.2 Business Domain

The system operates in the EdTech and Human Resource Technology (HR-Tech) domain, specifically at the intersection of skill assessment, career counselling, and workforce development. It serves educational institutions, vocational training centres, individual learners, and industry recruiters who require reliable, data-driven mechanisms to evaluate competencies and connect learners with suitable career opportunities.

## 1.3 Document Conventions

Requirements are identified using the prefixes FR (Functional Requirement) and NFR (Non-Functional Requirement), followed by a sequential number, for example FR-01 and NFR-01. Mandatory requirements are expressed using “shall”; desirable requirements are expressed using “should”.

## 1.4 Intended Audience

- Software developers and system architects responsible for building the platform
- Project evaluators and academic reviewers assessing the requirement analysis
- Educational institutions and training organizations adopting the platform
- Quality assurance and testing teams

# 2. Problem Description

Many students and job seekers struggle to identify careers that align with their skills and interests due to limited career guidance and generic, one-size-fits-all assessments. Traditional career counselling methods are largely manual, rely on the subjective judgement of a small number of counsellors, and often lack real-time labour market insights. As a result, learners frequently pursue training paths that are misaligned with actual industry demand, which lowers employability and increases the skills gap between education and employment.

The proposed AI-Powered Smart Vocational Skill Assessment and Career Recommendation Platform addresses this problem by using artificial intelligence and data analytics to assess a learner's technical and soft skills, analyze aptitude and learning preferences, compare the learner's competencies against current industry requirements, and recommend personalized career paths, certification programs, and job opportunities. In doing so, the system bridges the gap between education and employment through continuous, data-driven, and personalized guidance.

### 2.1 Stakeholders

- **Student / Job Seeker** — Primary end-user; takes assessments and receives career guidance
- **Trainer / Training Institution** — Manages courses and certification content; monitors learner performance
- **Industry / Recruiter** — Posts skill requirements and job openings; views matched candidates
- **System Administrator** — Manages users, roles, assessment content, and overall system health
- **Academic Institution Management** — Uses aggregated placement and employability data for accreditation and planning

### 2.2 Existing Challenges

- Career guidance is generic and not personalized to individual skill profiles
- No real-time comparison between learner competencies and industry demand
- Manual assessment processes are time-consuming and inconsistent
- Limited visibility into relevant certification and training options
- Absence of continuous progress tracking after initial assessment

### 2.3 System Objectives

The platform is designed around eight core objectives that together close the gap between learner potential and industry opportunity:


## 3. Scope of the System

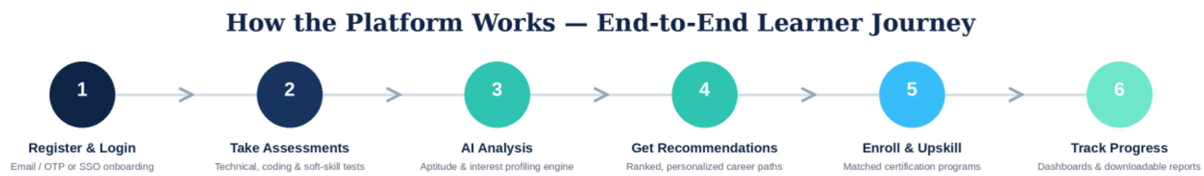
The system shall provide a web and mobile-accessible platform that allows learners to register, complete technical and soft-skill assessments, receive AI-generated career recommendations, browse and enroll in suggested certification programs, and track their learning progress over time. Training institutions shall be able to manage assessment content and monitor learner performance, while industry partners shall be able to post skill requirements and view matched candidate profiles. Administrators shall manage users, roles, and overall system configuration.

### 3.1 In Scope

- Technical and soft-skill assessment modules
- AI-based aptitude analysis and career path recommendation engine
- Skill-to-industry demand matching
- Certification and training program suggestions
- Personalized career report generation

### 3.2 Out of Scope

- Direct payment processing for third-party certification providers (handled via redirection to partner platforms)
- Full-fledged Learning Management System (LMS) content delivery (course videos, quizzes) beyond linking to partner platforms
- Physical placement / interview scheduling logistics




## 4. Functional Requirements

### 4. Functional Requirements

The functional requirements define the specific services and capabilities that the AI-Based Career Guidance and Skill Assessment Platform must provide to its users. These requirements focus on enabling learners to assess their skills, receive personalized career guidance, track progress, and interact with educational institutions and industry partners. The requirements are prioritized based on their importance to the system's core functionality.

FR-01: User Registration and Authentication (High Priority) : The system shall allow users to register and authenticate using email, OTP verification, or Single Sign-On (SSO) mechanisms. This functionality ensures secure access to platform services and user-specific data.

FR-02: Technical Skill Assessment (High Priority) : The system shall provide technical skill assessments, including multiple-choice questions (MCQs) and coding challenges, across various domains such as programming, data science, cybersecurity, and artificial intelligence.

FR-03: Soft-Skill and Psychometric Assessment (High Priority) : The system shall conduct soft-skill and psychometric assessments to evaluate communication abilities, teamwork, leadership potential, personality traits, and behavioral characteristics.

FR-04: AI-Based Aptitude and Learning Preference Analysis (High Priority) : The system shall utilize an Artificial Intelligence engine to analyze learner aptitude, interests, strengths, weaknesses, and learning preferences based on assessment outcomes.

FR-05: Personalized Career Path Recommendation (High Priority) : The system shall generate personalized career path recommendations by combining learner skill profiles, aptitude scores, interests, and market demand information.

FR-06: Skill Gap Analysis (High Priority) : The system shall compare learner competencies with current industry and labor-market requirements to identify existing skill gaps and areas for improvement.

FR-07: Certification and Training Recommendations (Medium Priority) : The system shall recommend suitable certification programs, online courses, workshops, and training opportunities that help learners address identified skill gaps.

FR-08: Career Report Generation (High Priority) : The system shall generate a comprehensive downloadable PDF report containing assessment results, skill analysis, career recommendations, and suggested learning pathways.

FR-09: Learning Progress Tracking (Medium Priority) : The system shall enable learners to monitor their learning activities, completed certifications, skill improvements, and overall career development progress over time.

FR-10: Trainer and Institution Management (Medium Priority) : The system shall allow trainers and educational institutions to manage assessments, learning materials, student performance records, and training programs.

FR-11: Industry Partner Integration (Medium Priority) : The system shall enable industry partners and employers to post job requirements and identify candidates whose skills and qualifications match organizational needs.

FR-12: Administrative Management (High Priority) : The system shall provide administrators with tools to manage users, roles, permissions, assessments, content, and overall platform operations.

FR-13: Notification Services (Low Priority) : The system shall send automated email and SMS notifications for assessment reminders, certification deadlines, learning recommendations, and important platform updates.

FR-14: Analytics and Reporting (Medium Priority) : The system shall generate analytical dashboards and reports that provide insights into learner performance, skill trends, assessment outcomes, and placement success rates

## 5. Non-Functional Requirements

The non-functional requirements specify the quality standards and operational characteristics that the AI-Based Career Guidance and Skill Assessment Platform must satisfy. These requirements ensure that the platform remains efficient, secure, scalable, reliable, and user-friendly.

NFR-01: Performance : The system shall generate AI-driven career recommendations within **5 seconds** under normal operating conditions to provide a responsive user experience.

NFR-02: Scalability : The platform shall support a minimum of 50,000 concurrent users without significant degradation in performance, ensuring future growth and wider adoption.

NFR-03: Security : The system shall secure data using TLS 1.2 or higher for data transmission and AES-256 encryption for data storage. Role-Based Access Control (RBAC) shall be implemented to restrict unauthorized access.

NFR-04: Reliability : The platform shall maintain at least 99.5% system uptime and include automated backup, recovery, and fault-tolerance mechanisms to minimize service disruptions.

NFR-05: Usability : The user interface shall be intuitive, user-friendly, and accessible to first-time users. The platform shall comply with WCAG 2.1 AA accessibility standards.

### NFR-06: Maintainability

The system shall adopt a modular and layered architecture that allows independent updates, testing, and maintenance of individual components without affecting the entire platform.

### NFR-07: Availability

Core platform services shall be available 24 hours a day, 7 days a week, except during scheduled maintenance periods.

### NFR-08: Portability

The platform shall function consistently across major web browsers and support both Android and iOS mobile devices to ensure broad accessibility.

### NFR-09: Interoperability

The system shall provide standardized RESTful APIs to facilitate integration with external job portals, certification providers, educational institutions, and third-party learning platforms.

## Quality Attributes of the Proposed System

The proposed architecture is designed to satisfy several critical quality attributes. Scalability ensures that the platform can accommodate a large number of users and increasing data volumes. Performance enables quick assessment processing and AI recommendation generation. Security protects sensitive learner information through encryption and access controls. Reliability and availability ensure uninterrupted access to services with minimal downtime. Maintainability supports future enhancements and simplified system updates. Usability improves user engagement through an intuitive interface, while interoperability enables seamless integration with external systems and services. These quality attributes collectively contribute to a robust, efficient, and future-ready AI-Based Career Guidance and Skill Assessment Platform.

# 6. Actors and Use Cases

This section identifies everyone and everything that interacts directly with the platform — the human roles who use it day to day, and the system-level actors that support it behind the scenes — and maps out the principal actions each of them is expected to perform.

## 6.1 Actors

Primary actors initiate interactions and drive the platform's core workflows, while secondary (system) actors operate in a supporting capacity, either processing data on the platform's behalf or supplying external content it depends on.

- **Student / Job Seeker (Primary) —** Registers, takes assessments, views recommendations, tracks progress
- **Trainer / Institution (Primary) —** Manages training/certification content and monitors learner performance
- **Industry / Recruiter (Primary) —** Posts skill requirements, views matched candidate profiles
- **System Administrator (Primary) —** Manages users, roles, assessment content, and system configuration
- **AI Engine (Secondary / System) —** Performs aptitude analysis, recommendation generation, and skill matching
- **Certification Provider API (Secondary / System) —** External service supplying course and certification data

### 6.2 Major Use Cases by Actor

The following use cases describe the interactions between different actors and the AI-Based Career Guidance and Skill Assessment Platform. Each actor performs specific tasks based on their role and responsibilities within the system. Together, these use cases cover the complete workflow of the platform, from learner registration and assessment to administrative management and industry collaboration.

#### 6.1 Student / Job Seeker

The **Student/Job Seeker** is the primary user of the platform. This actor utilizes the system to assess skills, receive career guidance, improve competencies, and monitor learning progress.

**Use Cases:**

- Register and Login to the platform.
- Take Technical Skill Assessments.
- Complete Aptitude and Psychometric Tests.
- View AI-Generated Career Recommendations.
- Enroll in Recommended Training Programs or Certifications.
- Download Personalized Career Reports.

#### 6.2 Trainer / Institution

The **Trainer/Institution** manages educational content and monitors learner development to ensure effective skill enhancement and career readiness.

**Use Cases:**

- Create and Manage Training Programs.
- Manage Certification Courses and Learning Resources.
- Monitor Learner Performance and Progress.
- Analyze Assessment Results.
- Track Course Completion Rates.

#### 6.3 Industry / Recruiter

The **Industry/Recruiter** represents employers and organizations seeking skilled candidates. This actor contributes industry requirements and evaluates suitable candidates.

**Use Cases:**

- Post Job Requirements and Skill Demands.
- View Matched Candidate Profiles.
- Search for Qualified Candidates.
- Review Candidate Skill and Assessment Reports.
- Identify Potential Recruitment Opportunities.

#### 6.4 System Administrator

The **System Administrator** oversees the overall operation, security, and maintenance of the platform.

**Use Cases:**

- Manage Users and Access Roles.
- Configure and Maintain Assessment Content.
- Monitor Platform Performance and Analytics.
- Generate Placement and Success Reports.

- Manage Industry Partnerships and Integrations.
- Monitor System Security and Compliance.
- Handle User Support and System Maintenance.

.

## 6.3 Use Case Diagram

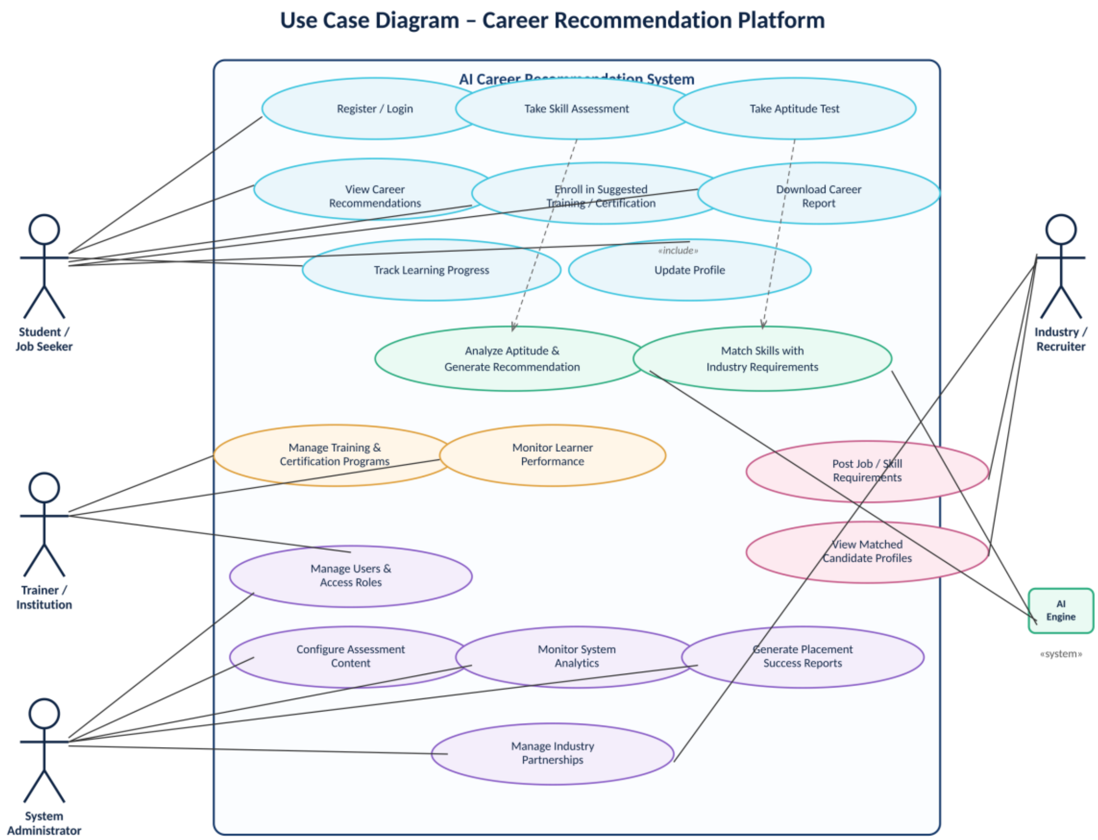



Figure 3: Use case diagram showing actors and their interactions with the system

## 6.4 Detailed Use Case Descriptions

### UC-01: Take Skill Assessment

**Primary Actor:** Student / Job Seeker

**Precondition:** Learner is registered and logged in.

**Main Flow:** 1) Learner selects a skill domain. 2) System presents technical questions/coding tasks. 3) Learner submits responses. 4) System scores the assessment and stores results.

**Postcondition:** Assessment results are saved and made available to the recommendation engine.

**UC-02: Analyze Aptitude and Generate Recommendation**

**Primary Actor:** AI Engine

**Precondition:** Skill assessment and aptitude test results are available.

**Main Flow:** 1) AI engine retrieves learner's assessment and aptitude data. 2) Engine runs the prediction model against industry demand data. 3) Engine ranks candidate career paths. 4) System displays top recommendations to the learner.

**Postcondition:** Personalized career recommendations are generated and stored for reporting.

**UC-03: Post Job / Skill Requirements**

**Primary Actor:** Industry / Recruiter

**Precondition:** Recruiter account is verified and active.

**Main Flow:** 1) Recruiter logs in to the industry portal. 2) Recruiter enters required skills and job details. 3) System stores the requirement and includes it in the skill-matching module.

**Postcondition:** Skill requirement is available for matching against learner profiles.

## 7. Data Requirements

The platform requires structured storage of learner profile data, assessment content and responses, AI model outputs, and industry demand data. The key data entities are summarized below.

- **Learner Profile —** User ID, name, contact, education, skills, interests. *(Source: Student registration)*
- **Assessment Response —** Assessment ID, question set, answers, score, timestamp. *(Source: Skill / aptitude tests)*
- **Recommendation Record —** Recommendation ID, learner ID, career paths, confidence score. *(Source: AI recommendation engine)*
- **Certification Catalogue —** Course ID, provider, skills covered, duration, prerequisites. *(Source: Certification provider API)*
- **Industry Requirement —** Job ID, required skills, experience level, recruiter ID. *(Source: Industry / recruiter portal)*
- **Progress Log —** Learner ID, activity type, completion status, date. *(Source: Platform activity tracking)*

## 7.1 Data Flow Overview

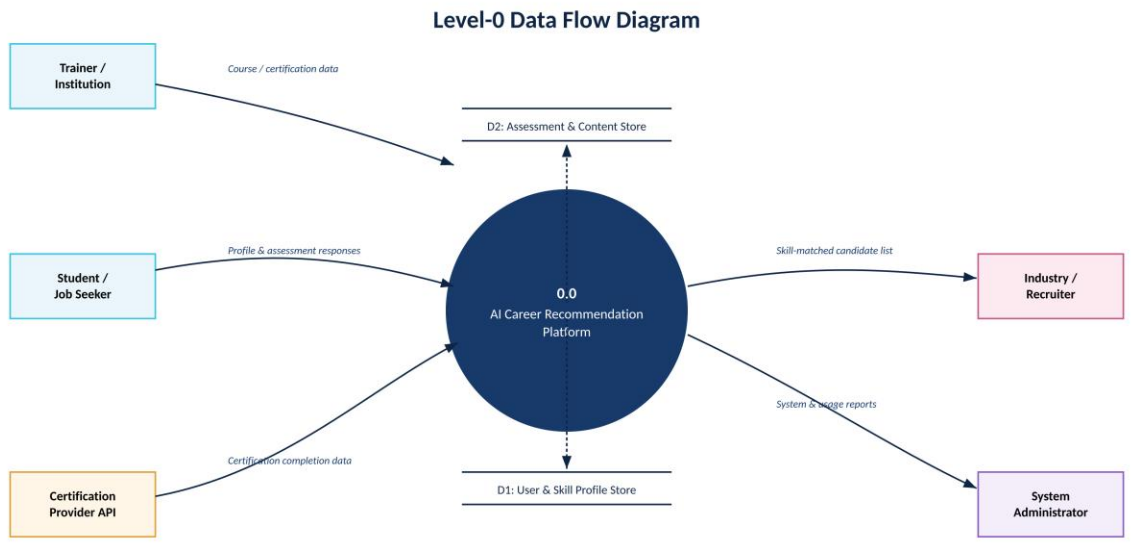



Figure 4: Level-0 DFD showing major data flows between external entities and the system

# 8. External Interface Requirements

## 8.1 User Interfaces

- Responsive web application accessible via desktop and tablet browsers
- Native mobile application for Android and iOS with offline-capable assessment caching
- Role-based dashboards for learners, trainers, recruiters, and administrators

## 8.2 Hardware Interfaces

- Standard client devices (desktop, laptop, smartphone) with internet connectivity
- Cloud server infrastructure with horizontal auto-scaling capability

## 8.3 Software Interfaces

- Integration with certification provider APIs (e.g., Coursera, NSDC) for course catalogue data
- Integration with job portal / labour-market data APIs for real-time industry demand
- Email and SMS gateway integration for notifications
- OAuth2 / SSO providers for authentication

## 8.4 Communication Interfaces

- RESTful APIs secured with HTTPS/TLS for all client-server and third-party communication
- Webhooks for asynchronous updates from certification provider platforms

# 9. System Features

The proposed AI-Based Career Guidance and Skill Assessment Platform is designed using a layered architecture that separates user interaction, business logic, security, integration, and data management functionalities. The system provides the following key features:

- User registration and secure authentication using Email, OTP, and SSO.
- Technical skill assessments through MCQs and coding challenges.
- Soft-skill and psychometric assessments for aptitude evaluation.
- AI-powered analysis of learner skills, interests, and learning preferences.
- Personalized career path recommendations based on assessment results.
- Skill gap analysis by comparing learner profiles with industry requirements.
- Recommendations for certifications, courses, and training programs.
- Downloadable PDF career reports with detailed insights and suggestions.
- Progress tracking dashboard to monitor learning and certification achievements.
- Analytics and reporting features for learners, institutions, and administrators.

**System Architecture – Layered View**

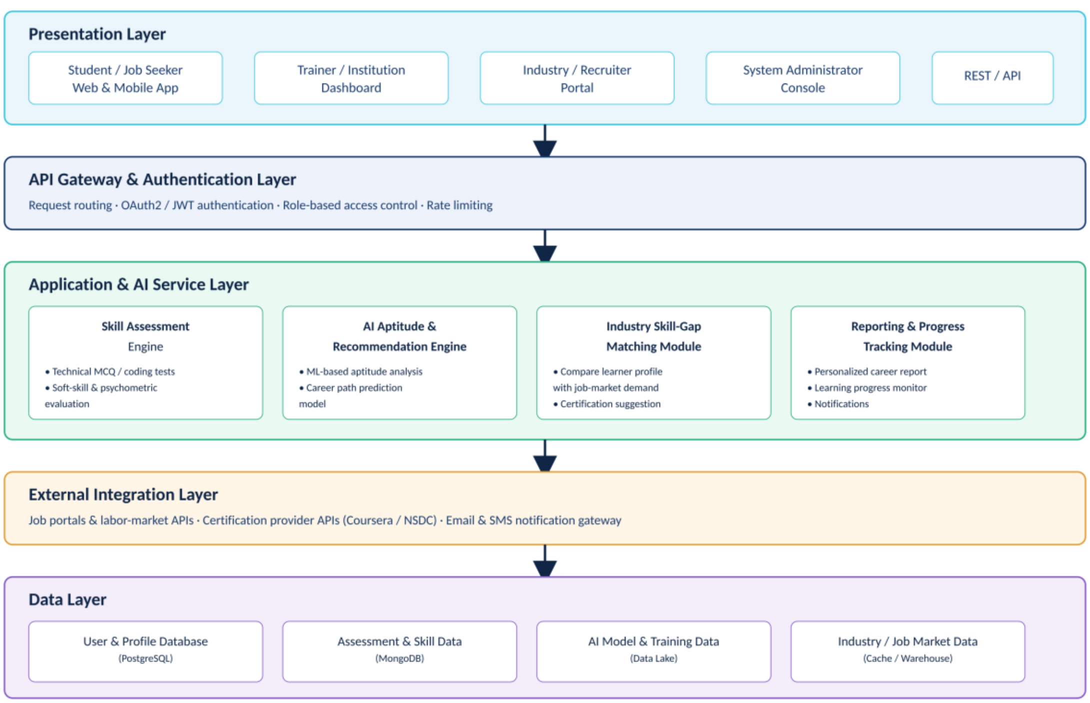



Figure 2: Layered architecture showing data flow from user interfaces through the AI service layer to persistent storage

### 9.1 Key Feature Modules

**Skill Assessment Engine**

Delivers technical MCQs, coding challenges, and soft-skill/psychometric evaluations, and produces normalized skill scores for each learner.

**AI Aptitude and Recommendation Engine**

Applies machine learning models to assessment and aptitude data to predict suitable career paths and rank them by fit and market relevance.

**Industry Skill-Gap Matching Module**

Continuously compares learner skill profiles against current industry and job-market requirements to identify skill gaps and suggest closing actions.

**Reporting and Progress Tracking Module**

Generates personalized, downloadable career reports and dashboards that track a learner's progress over time.

## 10. Assumptions and Constraints

### 10.1 Assumptions

- Learners have access to a device with a stable internet connection to complete assessments
- Industry requirement data provided by recruiters and third-party APIs is reasonably accurate and up to date
- Certification providers expose stable APIs for course catalogue synchronization
- Learners provide truthful information during registration and assessment
- A sufficient volume of historical assessment and placement data will be available to train and validate the AI recommendation model

### 10.2 Constraints

- **Technical —** The AI recommendation model must be retrainable without downtime to the core assessment service.
- **Operational —** The system must support institutional onboarding without requiring changes to existing student information systems.
- **Economic —** Development and cloud infrastructure costs must remain within the allocated academic project / institutional budget.
- **Legal —** The system must comply with applicable data protection regulations (e.g., IT Act, GDPR where applicable) regarding storage of personal and assessment data.
- **Time —** The system must be delivered within the academic project timeline defined for the course.

# 11. Requirement Validation — Completeness and Consistency

## 11.1 Stakeholder Coverage Check

Each stakeholder group identified in Section 2.1 has at least one corresponding functional requirement and use case: learners are covered by FR-01 through FR-09, trainers by FR-10, recruiters by FR-11, and administrators by FR-12 through FR-14. This confirms that all identified stakeholder needs have been addressed in the requirement set.

## 11.2 Ambiguity, Redundancy, and Consistency Review

**Ambiguity**

**Finding:** Initial draft of FR-05 did not specify what “personalized” recommendation means.

**Resolution:** Clarified to reference skill, aptitude, and interest data explicitly.

**Redundancy**

**Finding:** Progress tracking initially appeared under both learner and reporting features.

**Resolution:** Consolidated into a single Reporting and Progress Tracking module (Section 9.1) **Inconsistency.**

**Finding:** NFR draft implied both 99.9% and 99.5% uptime in different notes.

**Resolution:** Standardized to 99.5% uptime (NFR-04) to align with realistic cloud SLA levels.

**Missing Requirement**

**Finding:** Notification preferences were not addressed in the initial pass.

**Resolution:** Added FR-13 covering email/SMS notifications.

## 11.3 Suggested Improvements

- Introduce a formal requirement traceability matrix linking each FR/NFR to its corresponding use case and test case.
- Add explicit data-retention and consent-management requirements to strengthen legal compliance.
- Define measurable acceptance criteria (e.g., recommendation accuracy threshold) for the AI recommendation engine.
- Conduct periodic requirement review sessions with representative stakeholders (students, trainers, recruiters) as the project evolves.

# 12. Conclusion

This Software Requirement Specification document has analyzed the scenario for the AI-Powered Smart Vocational Skill Assessment and Career Recommendation Platform, identified its stakeholders and system objectives, and defined a comprehensive set of functional and non-functional requirements. The actors and use cases were modeled to reflect real interactions between learners, institutions, recruiters, administrators, and the AI engine, supported by a layered system architecture and data flow model. Assumptions and constraints were documented, and the requirement set was validated for completeness, consistency, and ambiguity. This document provides a solid foundation for subsequent design and implementation phases of the project.